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EXTENDED ABSTRACT

Evaluating ArchesWeather and ArchesWeatherGen under Multi-Decadal AMIP-style climate simulations

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1. Introduction and objectives

Data-driven models trained for weather forecasting on ERA5 [Hersbach et al.(2020)] now surpass the skill of numerical models at a fraction of the computational costs [Bi et al.(2023), Lam et al.(2023)]. Furthermore, stochastic modeling approaches, such as diffusion models, provide a natural avenue to incorporate the uncertainty in earth system dynamics [Price et al.(2025), Alet, et al.(2025)]. Although still in their infancy compared to weather models, data-driven methods are also advancing long-term climate emulation. Current models are, however, either deterministic, *e.g.*, [Watt-Meyer et al.(2025), Guan et al.(2025)], or are trained on numerical model simulations, *e.g.*, [Clyne et al.(2025)]. Here, we evaluate a stochastic modeling approach, ArchesWeatherGen [Couairon et al.(2025)], on historical climate as well as its response to forcings. Our setup resembles the Atmospheric Model Intercomparison Project protocol (AMIP), part of the Coupled Model Intercomparison Project (CMIP6), where the atmospheric model is constrained with prescribed sea surface temperature and sea ice. These simulations contribute to AIMIP (AI Model Intercomparison Project) [Henn et al.(2025)], an initiative to compare current state-of-the-art AI climate models.

2. Methods and data

ArchesWeather and ArchesWeatherGen are data-driven models designed for medium-range weather forecasting. ArchesWeather is a deterministic model, whereas ArchesWeatherGen is a probabilistic generative model based on flow matching, which corrects ArchesWeather's predictions and captures the temporal variability of the system. Both models share the same transformer backbone, which improves upon the Pangu-Weather architecture with increased efficiency.

Following AIMIP protocol, we adapt the models to serve as atmospheric climate models by extending them to take into account monthly mean forcings for sea surface temperature (SST) and sea ice (SIC). Models are trained from scratch on daily averaged 1º ERA5 data from 1979–2013 and forecast the atmospheric state at a 24-hour lead time conditioned on the previous two time steps.

3. Preliminary results and/or expected outcomes

We examine the ability of both ArchesWeather and ArchesWeatherGen to emulate the current climate by comparing them to the ERA5 climatology on both train (1979–2013) and test (2015–2024) periods. We run simulations from October 1978 to December 2024, starting from an initial condition from ERA5 (1 Oct. 1978) and auto-regressively feeding back the model predictions as input for subsequent predictions. ArchesWeatherGen generates a stable annual trend (Figure 1a), matches the time mean of

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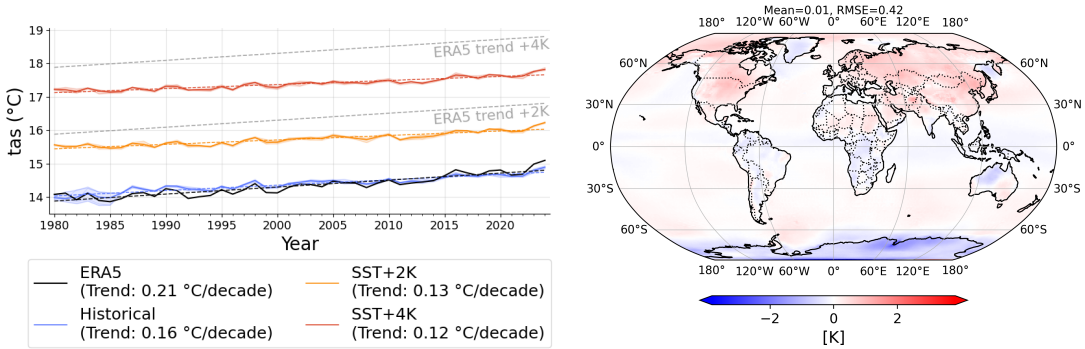
ERA5 (Figure 1b), reproduces teleconnections, and shows annual cycles of variability (e.g., the Northern Annular Mode). We further examine the models' response to forcings by evaluating the stability, trend, and physical correlations under different forcing scenarios. Figure 1a showcases the surface temperature shift in response to increased SST. We ablate different model configurations and specifically investigate ArchesWeatherGen compared to the deterministic version, ArchesWeather. We also analyze the models' capability to reproduce extreme weather statistics (*e.g.* return periods for extreme heat waves).



(a) (b)

Figure 1. *Surface temperature predictions from ArchesWeatherGen 5-member ensemble mean compared against ERA5 ground truth. (a) When forced with historical SST/SIC (blue), the model's annual global mean trend reproduces the historical trend of ERA5 ground truth (black). When SST forcing is uniformly increased by +2 K (orange) and +4 K (red), the model responds with shifted temperature trends. (b) Climatological bias between ArchesWeatherGen ensemble mean and ERA5, showing larger differences over land surface than over ocean.*

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